Liem Fairbairn

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EDUCATION

Imperial College London — MEng Aeronautical Engineering (4-year Integrated Master's) 2025 - 2029
Year 1 Final Grade: 68.7% (2:1)

Alexandra Park School 2018 - 2025
A Levels: Mathematics (A*), Physics (A*), Classical Civilisation (A*), Further Mathematics (A)

TECHNICAL SKILLS

Programming (MATLAB, Python, C++), CATIA via Dassault 3DEXPERIENCE (Part Design, Assembly, Drafting), KiCAD (PCB Design), STAR-CCM+ (CFD), Abaqus (FEA), Experimental Testing & Data Analysis, Technical Report Writing

FEATURED PROJECT

NASA TechLeap Robotically Manipulated Payload Challenge - Independent Submission 2026

- Developed an engineering proposal for an in-space electric propulsion propellant (xenon) transfer payload, independently authoring a full thermodynamics/structural systems requirement addressing NASA propulsion shortfall 23.07.
- Modelled supercritical blowdown dynamics using a real-gas Redlich-Kwong equation of state in MATLAB for verification and verified dual pressure vessel margins (6061-T6 aluminium, 42 MPa worst-case design pressure), for SF of 3 and 0.71kg of Xenon.
- Designed a motorised valve control and electromagnetic docking/locking subsystems for autonomous fluid transfer via a Free-Flying Robot arm, with a conceptualized microcontroller-based sensor system and data-logging.
- Produced CAD concept assembly (Dassault 3DEXPERIENCE), cost/budget narrative, engineering drawings, and video pitch as sole engineer and systems owner on the submission.

ENGINEERING PROJECTS & EXPERIMENTAL TESTING

2 Blade Wind Turbine Design and Manufacturing 2026

- Designed and manufactured two turbine blades in a team of 6, extracting 13.9 W in a 10 m/s wind tunnel at 1600 rpm.
- Developed MATLAB Blade Element Momentum solver with aerodynamic correction models to predict and validate performance, down selecting aerofoil configurations.
- Evaluated factors of material, cost, time, manufacturing, aerodynamic, and structural properties to develop multi-material (PLA, Carbon fibre, Plywood, and foam) final design.

Custom Drone Flight Controller Design (Ongoing) 2026 - Present

- Designing custom STM32F411-based flight controller integrating an ICM-46270-p IMU, MLX90394 magnetometer and BMP390 barometer for inertial, magnetic, and altitude sensing.
- Designed electronics architecture and KiCAD Schematic/PCB including power regulation and decoupling, SPI and I2C communications and sensor interrupt interfaces.

Structures and Materials Laboratory Work 2025 - 2026

- Built and mesh-converged 1D and 3D Abaqus FE models of statically indeterminate steel beam under 40N load. Validated against Euler-Bernoulli theory and physical strain gauge testing Achieved convergence to 4mm at 28 elements
- Tensile tested aluminium and polymer specimens to fracture, identifying Young's Modulus and deformation behaviour.

Aerodynamics and Aircraft Design Laboratory Work 2025 - 2026

- Built CFD model (STAR-CCM+) of laminar pipe flow to assess validation against experimental and Hagen-Poiseuille theory, achieving 3.71% agreement in pressure-coefficient gradient.
- Derived stability and control derivatives (neutral point, downwash gradient $d\varepsilon/d\alpha = 0.96$, tailplane lift curve slope = 0.25/degree) of aircraft model via wind tunnel and flight simulation testing across 3 tailplane settings confirming a stable pitch break at sta

LEADERSHIP & COMPETITIONS

M3 Mathematics Modelling Challenge – Thermal Modelling 2025

- Led a seven-person team, developing Python-based simulations of thermal behaviour in buildings and infrastructure resilience under power outage scenarios.

ISSET Mission Discovery Payload Conception 2024

- Collaborated in a team of 6 to design a microgravity-centred experiment assessing changes in the material properties of stringlike materials (spider silk) for medical applications.
- Designed an electromagnetic tensile testing mechanism against dimensional and mass constraints.